BHUVANESHWARI M.C.

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PROFESSIONAL SUMMARY

Nanotechnology engineer (M.Sc., KTH Royal Institute of Technology) with hands-on cleanroom fabrication experience (PECVD, ALD, photolithography, RIE) and a strong foundation in compound semiconductor device physics, including III-V HEMTs, 2DEG transport, and bandgap engineering. Skilled in electrical device characterization (IV/CV parameter extraction) and TCAD-based device modelling. Seeking to apply this fabrication-to-modelling skill set to wide-bandgap and ultra-wide-bandgap semiconductor devices for RF and power electronics.

RESEARCH INTEREST

Wide-Bandgap & Ultra-Wide-Bandgap HEMTs, III-Nitride Device Fabrication, Semiconductor Device Characterization, Compact & TCAD Modelling, RF and Power Electronics

RESEARCH EXPERIENCE

Master Thesis Student - Si Quantum Dot Optical Characterization **Feb 2023 – Feb 2024**

KTH Royal Institute of Technology, Stockholm, Sweden

- Investigated High Pressure Water Vapour Annealing effects on individual Si quantum dots using micro-PL spectroscopy; improved emission stability by 2.5× at 2.6 MPa annealing pressure.
- Resolved single-dot PL spectra (24- 39 nm FWHM, 630- 676 nm) and applied statistical threshold analysis to quantify blinking ON/OFF frequency - 0.04 Hz reduction observed.

Code: KTH-Projects/Master Thesis

Research Intern - Sensors & Instrumentation **May 2019 – Jun 2019**

Indian Space Research Organisation (ISRO), India

- Improved sensor calibration accuracy by 20% using LabVIEW-based instrumentation workflows; automated data acquisition and hardware interfacing procedures.

CLEANROOM & DEVICE FABRICATION EXPERIENCE

- Performed PECVD, ALD, photolithography, and RIE processing on SOI CMOS structures, fabricating 0.96 μm^2 contact holes and analysing process-induced defects.

Code: KTH-Projects/Nanofabrication/CMOS.md

- Reduced AFM roughness measurement uncertainty from 41 nm to 14.74 nm through scan parameter optimization.
- Performed surface morphology and elemental analysis using AFM and SEM to support process and material quality assessment

Code: KTH-Projects/Characterization

INDEPENDENT RESEARCH PROJECTS

Self-driven modelling of AlGaN/GaN HEMT polarization charge and band offsets, grounded in AlGaAs/GaAs heterojunction fundamentals -bandgap, lattice mismatch, and band alignments; additional projects in progress.

Code: Bhuvana15598/Semiconductor-Projects

DEVICE CHARACTERIZATION & MODELLING

- Extracted ideality factor (n = 3.90), series resistance (6.7 Ω), and doping concentration (4.69×10^{13} cm^{-2}) via SPICE-based IV/CV fitting; investigated non-linear transport behaviour across 298–383 K.
- Simulated SOI and bulk MOSFETs across 22–90 nm nodes using PADRE/Prophet TCAD, with Python/MATLAB-based post-processing; validated Id–Vg curves within 5% agreement and extracted DIBL and Vt roll-off.
- Coursework in HEMT physics, 2DEG transport, III-V heterostructures, and bandgap engineering.

EDUCATION

M.Sc. Nanotechnology - KTH Royal Institute of Technology **Aug 2021 – Apr 2026**

Stockholm, Sweden

Coursework: Solid State Physics, Nanofabrication Technologies, Methods and Instruments of Analysis, Microsystem Technologies, Design of Nano Semiconductor Devices, Quantum Materials and Devices, Semiconductor Devices, Compound Semiconductors.

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B.E. Electronics and Communication Engineering - Anna University **Aug 2016 – May 2020**

India

- Bachelor Thesis cum Publication, NCCS / Springer Journal (Jun 2020): co-authored research on delay line/TDC architecture, achieving 9.76 ps precision using Xilinx ISE.

SKILLS

Fabrication Processes: Photolithography, PECVD, ALD, RIE, SOI CMOS process flows

Characterization Tools: IV/CV, SEM, AFM, Photoluminescence Spectroscopy, Optical Microscopy

Semiconductor Physics: III-V Heterostructures, HEMT Physics, 2DEG, Bandgap Engineering

Simulation & Design Software: PADRE/Prophet (TCAD), COMSOL Multiphysics, LabVIEW, EasyEDA, NanoHub

Programming Languages: Python, MATLAB

Data Analysis Tools: GitHub, MATLAB, ImageJ, OriginPro

ACHIEVEMENTS

- Distinction, CeNSE Winter School, Indian Institute of Science, Bangalore.
- Semi-Finalist, Texas Instruments Innovation Challenge - prototyped "SMARTKART", an automated IoT-integrated shop/scan/go system.

REFERENCES

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